# ADR-114 - Search Endpoint Information by Service ID

Find the Right Service

6 May, 2026

# Table of Contents

| | |
|---|---|
| ADR Number | ADR-114 |
| Name | Search Endpoint Information by Service ID |
| Status | REVIEW |
| Decision | TBC |
| Stakeholders | Architects, BAs, Tech Leads, Engineers |

# 1 Overview

Today, many connecting parties search by Service ID after a Searching by Triage code in order to make an onwards referral to a service.

Having discussed with connecting parties as part of our early analysis on how they are using these endpoints, we confirmed that this would still be needed in a replatformed Directory of Services because:

- Although the endpoints for each service will be available in Search by Triage Code, some connecting parties cannot persist this data in order to make a service referral later
- BaRs do not yet have dates where they would offer this onwards Referral and Booking Service to connecting parties, so we currently do not have a deprecation date for offering endpoint information

The Directory of Services Search API [1]currently supports the following endpoints:

- Production ready:
  - Retrieval of Organisation details by ODS code, including associated Endpoints
- In development:
  - Retrieval of HealthcareService resources by ODS code, optionally including providing Organisation, Endpoint and Location resources
  - Retrieval of HealthcareService resources by Triage code

This ADR defines the design for an additional capability:

- **Retrieval of Endpoint information by Service Identifier (UUID)**

A separate ADR addresses:

- Retrieval of HealthcareService information by Service Identifier (UUID)[2]

These ADRs are being developed in parallel as the design decisions are interdependent.

1. https://digital.nhs.uk/developer/api-catalogue/directory-of-services-search-api
2. https://nhsd-confluence.digital.nhs.uk/spaces/FRS/pages/1352225259/ADR-113+-+DRAFT+Search+HealthcareService+by+Service+ID

# 2 Drivers

The design must support retrieval of Endpoint information by service identifier, while ensuring:

- Alignment with FHIR R4 standards and existing DoS API patterns
- Consistency with existing endpoints:
  - GET /HealthcareService?organization.identifier=https://fhir.nhs.uk/Id/ods-organization-code|ABC123[3]
    - Supports _include for Endpoint, Location and Organisation
  - GET /Organization?identifier=https://fhir.nhs.uk/Id/ods-organization-code|ABC123[4]
    - Supports _revinclude=endpoint:organization
- Possible functionality deprecation following BaRS implementation

**Business requirements**

- Retrieve Endpoint resources linked to a specific HealthcareService
- Minimise number of API calls required by consumers

**Architectural considerations:**

- Adherence to standard FHIR interaction patterns (search vs read)
- Clear and consistent API design
- Minimise risk of introducing duplicate API capability that may later overlap with future BaRS API

3. https://fhir.nhs.uk/Id/ods-organization-code%7CABC123
4. https://fhir.nhs.uk/Id/ods-organization-code%7CABC123

# 3 Assumptions

- Service ID maps to FHIR logical id of HealthcareService - confirmed
- Endpoint relationships are modelled via HealthcareService.endpoint references
- Future BaRSAPI may provide equivalent or overlapping endpoint information
- Delivery timelines for the future capability are not currently confirmed

# 4 Decision

TBC

# 5 Options considered

## 5.1 Option 1 – FHIR include on existing HealthcareService search endpoint

```
GET /HealthcareService?_id={serviceId}&_include=HealthcareService:endpoint
```

**Example:**

```
Request: (existing endpoint)
GET /HealthcareService?_id=9f2c6f12-1a6d-4d9c-
a111-123456789abc&_include=HealthcareService:endpoint
Accept: application/fhir+json

Parameters:
_id = serviceId (new param)
_include=HealthcareService:endpoint (existing functionality)

Response:
 - FHIR Bundle (searchset) containing
    - 1 HealthcareService
    - 0..* Endpoint
```

**Pro:**

- Fully aligned with FHIR search and _include semantics
- Reuses existing /HealthcareService endpoint (no additional endpoint required)
- Consistent with existing API patterns (Organisation + _revinclude)
- Supports retrieval of related resources in a single request

**Con:**

- Returns mixed resource types within a Bundle (however existing endpoints work the same way)
- Requires consumers to extract Endpoint resources

**Risk:**

- Increased client-side parsing complexity
- Potential over-fetching of HealthcareService resource when only Endpoint is required

**Cost:**

- Low

## 5.2 Option 2 – Sub-resource style endpoint

```
GET /HealthcareService/{serviceId}/Endpoint
```

**Example:**

```
Request: (new endpoint)
GET /HealthcareService/9f2c6f12-1a6d-4d9c-a111-123456789abc/Endpoint
Accept: application/fhir+json

Parameters:
 - serviceId (new param)

Response:
 - FHIR Bundle (searchset) containing
   - 0..* Endpoint
```

**Pro:**

- Very intuitive and easy to consume
- Returns only Endpoint data in a bundle to accomodate multiple endpoints for each service

**Con:**

- Not a standard FHIR interaction

**Risk:**

- Introduces non-FHIR API pattern
- Harder to maintain consistency across APIs

**Cost:**

- Medium

## 5.3 Option 3 - Search parameter

FHIR Endpoint[5] does not contain a direct reference to HealthcareService. This means there is no standard service search parameter for HealthcareService search on Endpoint

However there is a relationship where HealthcareService[6] references Endpoint. Hence we have 2 options,

5. https://hl7.org/fhir/R4/endpoint.html
6. https://hl7.org/fhir/R4/healthcareservice.html

### 5.3.1 Option 3A — Custom Endpoint search parameter

FHIR allows defining a custom search parameter - "Servers are not required to implement any of the standard search parameters. Servers may also define their own parameters."[7]- FHIR R4

```
GET /Endpoint?serviceId={serviceId}
```

**Example:**

```
Request: (new endpoint)
GET /Endpoint?serviceId=9f2c6f12-1a6d-4d9c-a111-123456789abc
Accept: application/fhir+json

Parameters:
 - serviceId = custom server-defined search parameter for HealthcareService id

Response:
 - FHIR Bundle (searchset) containing
   - 0..* Endpoint
```

**Pro:**

- Aligns with FHIR search model (resource-centric search)
- Returns only Endpoint resources
- Could scale well for future endpoint-based queries

**Con:**

- Not a standard FHIR search parameter
- Requires custom server-defined search behaviour

**Risk:**

- Reduced interoperability with other FHIR implementations
- Additional documentation and governance required to explain custom behaviour
- Potential future rework if stricter FHIR conformance is required

**Cost:**

- Medium

### 5.3.2 Option 3B — Reverse chaining using _has

FHIR supports reverse chaining via _has, which is specifically for selecting resources based on other resources that refer to them.

7. https://hl7.org/fhir/R4/search.html#parameters

The spec describes _has as support for "reverse chaining - that is, selecting resources based on the properties of resources that refer to them[8]" - FHIR R4

```
GET /Endpoint?_has:HealthcareService:endpoint:_id={serviceId}
```

**Example:**

```
Request: (new endpoint)
GET /Endpoint?_has:HealthcareService:endpoint:_id=9f2c6f12-1a6d-4d9c-a111-123456789abc
Accept: application/fhir+json

Parameters:
 - _has:HealthcareService:endpoint:_id

Response:
 - FHIR Bundle (searchset) containing
   - 0..* Endpoint
```

**Pros:**

- FHIR-native approach using reverse chaining
- Returns only Endpoint resources
- Better standards alignment than a custom service parameter

**Cons:**

- More complex and less intuitive for consumers
- Harder to document and support operationally
- Depends on server support for _has, which is optional

**Risks:**

- Increased implementation complexity
- Increased consumer confusion due to advanced FHIR syntax

**Cost:**
Medium to High

## 5.4 Option 4 – Endpoint read by Endpoint ID

```
GET /Endpoint/{endpointId}
```

**Example:**

8. https://hl7.org/fhir/R4/search.html#has

```
Request: (new endpoint)
GET /Endpoint/9f2c6f12-1a6d-4d9c-a111-123456789abc
Accept: application/fhir+json

Parameters:
 - endpointId - id of endpoint resource

Response:
 - 1 Endpoint (not bundle)
```

**Pros:**

- Standard FHIR read interaction
- Simple and intuitive for consumers
- Returns only the requested Endpoint resource
- Useful where the Endpoint identifier is already known

**Cons:**

- Does not return all Endpoints associated with a HealthcareService
- Does not support retrieval by Service Identifier
- Does not meet the primary business requirement on its own
- DB model doesn't currently allow easy retrival of endpoints by id

**Risks:**

- Requires call chaining, as consumers must first determine which Endpoint identifiers are linked to the HealthcareService before retrieving them individually
- Could result in a high volume of API calls where a service has multiple Endpoints

**Cost:**
Low

## 5.5 Option 5 – Combined approach?

# 6 Rationale

## 6.1 Criteria

- FHIR compliance – alignment with standard FHIR R4 interaction patterns and avoidance of custom behaviour
- Meets user need – ability to retrieve Endpoint resources linked to a HealthcareService with minimal API calls
- Consistency – alignment with existing DoS Search API patterns
- Usability – ease of use and clarity for connecting parties
- Implementation effort – relative complexity to design, build, test and document
- Future convergence / deprecation effort – effort required to rationalise, migrate consumers, or deprecate the capability in future

## 6.2 Scoring

| Option | FHIR compliance | Meets user need | Consistency | Usability | Implementenattion Effort | Deprecation effort |
|---|---|---|---|---|---|---|
| Option 1 - FHIR include on HealthcareService search | High | Medium | High | High | Low | Low |
| Option 2 - Sub-resource style endpoint | Low | High | Low | High | Medium | Medium |
| Option 3A - Custom Endpoint search parameter | Medium | High | Medium | High | Medium | Medium |
| Option 3B - Reverse chaining using _has | High | High | Medium | Low | Medium | Medium |
| Option 4 - Endpoint read by Endpoint ID | High | Low | High | High | Medium | Medium |